

Look at it another way. You share a couple of Linux (of course!) PCs at home with family members with diverse requirements: your grandmother wants simplicity, the cousin wants multimedia, the kids want games and education tools, and you want to try out the latest distributions. Now you can meet all these requirements and pamper yourself in the bargain.

Let us get straight into looking at a partition and multi-boot scheme that you can use as-is or tailor to your requirements. I'm assuming that you understand the basics of partitioning, boot loaders, and have installed Linux distributions in the past. Even if you haven't, I am sure you can get a leg-up from the neighbourhood know-all.

Template

Our partition and multi-boot scheme has a two-level boot menu. It has a master boot menu listing all the installed distributions available for booting. Once you select the distro to boot from, you are taken to the dedicated GRUB menu of the specific distro you are booting.

For instance, when I select 'openSUSE 11' on the master boot menu, I am taken to the SUSE boot menu that lists the various options to boot openSUSE such as: a 'normal boot' or a 'fail-safe session'. We will revisit boot menus later in this article. For the moment let us get down to partitioning our disk. We are assuming a blank hard disk to keep things uncomplicated. You can start the partitioning session from a live CD such as Knoppix.

Start with a primary partition (sda1) of about 100 MB for */boot* followed by a primary partition (sda2) for swap of a suitable size. Make sure the swap is larger than your RAM because this is the space used by the suspend-to-disk feature on some distros.

Of the remaining two primary partitions possible, make one an extended partition. This extended partition (sda3) will house the following logical partitions:

- A partition (sda5) to mount as a common */tmp* directory for all distributions. About 2 GB for */tmp* should do for most people, but again this is a suggestion and not a recommendation.
- Slots (sda6, sda7, and so on) for the various distributions. About 10-12 GB per distribution should be adequate for most distributions and you could even do with less.

The fourth primary partition (sda4) can be mounted as the common */home* directory for all distributions. A common */home* across distributions can lead to some irritations related to the desktop and we will explore a workaround later when discussing usage scenarios. If you have a capacious hard disk (and who doesn't these days), you can have more than one partition available for mounting as */home* and specify which one to mount automatically for a particular distribution during install time. If you have more than one partition for */home*, make sure you create the additional ones as logical

Partitions and mount points						
Distro	/boot	/tmp	Swap	/	/home	GRUB
1st Distro	sda1(F)	sda5(F)	sda2(F)	sda6(F)	sda4(F)	In MBR
2nd Distro		sda5	sda2	sda7(F)	sda4	In sda7

Table 2

partitions inside the extended partition sda3. Choosing a common swap and */tmp* partition across distributions helps you utilise your hard disk real estate that much better. You have a choice of filesystem types to use for each partition, but remember that GRUB is finicky and prefers ext3.

Depending on how much space you need for swap, */tmp* and */home*, you can easily arrive at how many distributions you can pack into a single hard disk. As a variation you can also pack all distributions on one hard disk and keep your data on another.

Table 1 shows how the partitions might look like for a single disk of 120 GB with three distributions sharing a common home partition.

Install procedure

One approach is to start with a bootable GRUB disk and install GRUB in the MBR of sda. This way the boot partition is not associated with any distribution. But let us take a simpler alternative and let the first distro install the boot loader for us. This way, the dedicated */boot* partition gets associated with the first distribution but that does not make any material difference to the boot interface.

To keep things systematic and under control, start by installing the first distribution. Using Table 2 as a cue, install the first distro by providing the indicated mount points for the respective partitions. For this distro you will install GRUB in the MBR and use sda1 as */boot*. For all other distros, GRUB goes into the "first sector of the boot partition". The (F) next to each partition indicates it needs to be formatted. Note carefully that from the second distro onwards, you need to format only the "/" partition. Remember this, lest you end up losing data that already exists in the */home* or */tmp* partitions.

Configuring the GRUB menu

Since we are letting the first distro take charge of the boot partition and install GRUB for us, all we have to do is edit the GRUB menu to add the other distros. The GRUB menu exists as */boot/grub/menu.lst*. Here is what mine looks like today for four distros:

```
# grub.conf generated by anaconda
#
# Note that you do not have to rerun grub after making changes to this file
# NOTICE: You have a /boot partition. This means that
# all kernel and initrd paths are relative to /boot/, eg.
# root (hd0,0)
# kernel /vmlinuz-version ro root=/dev/sda6
```